



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT/2023-24/EMDSYD/867771

Date:

05/02/2024(10:08:35)

Subject:Review of LMI for Cathodic protection Of Sweet water and Sea water pipe lines

Cathodic Protection is provided for the sweet water and sea water pipelines for protecting the underground pipelines from soil corrosion. Accordingly, a site LMI has been prepared to regularly monitor and ensure the healthiness of the cathodic protection of Stage-1&2

LMI's are reviewed periodically for further improvement and incorporation of recommendation by COS if any. No chages were envisaged by COS.

Submitted for kind approval of competent authority please

Submitted by:

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Note No:2

The LMI is a site prepared LMI which doesn't have any corporate documentation system.

As per technical audit committee recommendations, OFS has to cross check the readings of the table contained in the LMI.

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Note No:5

may please be approved

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Note No:6

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NTPC Limited

SIMHADRI SUPER THERMAL POWER STATION

LMI/STN/ELEC/025
ISSUE NO: 1, REVISION: 4
DATE: JAN 2024

**CATHODIC PROTECTION
FOR
SEA WATER & SWEET WATER PIPELINES**

APPROVED FOR

IMPLEMENTATION BY - - - - -

CHIEF GENERAL MANAGER (SIMHADRI)

Date: _____

NTPC Limited
SIMHADRI SUPER THERMAL POWER STATION
OPERATION & MAINTENANCE DEPARTMENT

LMI (Name): CATHODIC PROTECTION FOR SEA WATER & SWEET WATER PIPELINES			
LMI No. : LMI /STN/ELEC/025			
Issue No.	1	Revision: 4	Date: Jan 2024

	Name	Designation	Signature
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CATHODIC PROTECTION FOR SEA WATER & SWEET WATER PIPELINES

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1.0 **OBJECTIVE: -**

Cathodic Protection is provided for the sweet water and sea water pipelines for protecting the underground pipelines from soil corrosion. Cathodic protection system supplements the protection offered by the coating and takes care of the coating defects that occur either during the lowering of the pipeline or in the course of its service.

The objective of this LMI is to regularly monitor and ensure the healthiness of the cathodic protection of Stage-1&2

2.0 **SCOPE:**

This LMI document covers cathodic protection of Stage-1 & 2 provided for

- Sweet Water Pipeline running from Sweet water pump house to the raw water reservoir. The pipeline is of mild steel and is given coal tar enamel for protection against corrosive action of soil.
- Sea water pipelines running from the sea water Pump House to main plant which is approximately 5Kms. The sea water pipelines comprises of one number pipeline to carry effluent discharge and One number each for sea water supply and return line.

3.0 **REFERENCE DOCUMENT:**

Stage-1 O&M Manual of M/s Cathodic Protection Pvt Ltd (OEM).

Stage-2 O&M Manual of M/s Metcor (OEM).

4.0 **SUPERSEDED DOCUMENTS:**

LMI /OGN/ELEC/025 Issue: 1 Rev: 3 Dated: Jan 2022

5.0 **SPECIFIC INSTRUCTIONS TO BE FOLLOWED**

INSTRUCTION	RESPONSIBILITY	DOCUMENT REF.
Monthly pipe to soil potential Readings to be measured and recorded	Sectional Head of EM of Makeup system	LMI/025

6.0 METHODOLOGY OF IMPLEMENTATION:

- Monthly readings of potential difference between Ground and Pipeline at all the test lead points (TLPs) will be taken with the help of anode cell.
- Pipe to Soil Potential (PSP), DC output voltage and output current readings to be recorded in the format given at page no.3.
- Verify that all the Pipe to Soil Potential (PSP) readings are within the acceptable limits of -0.850 V TO -1.4 V.
- If the Pipe to Soil Potential (PSP) readings are not within the range of -0.85 V TO -1.4 V, adjust internal set potential at TR station to bring the potential to acceptable limits.

7.0 RESPONSIBILITY CENTRE FOR IMPLEMENTATION:

Area engineer of electrical maintenance for makeup water system is responsible for implementation and overall responsibility lies with sectional head of electrical maintenance.

8.0 EXCEPTION REPORTING AND DISPENSATION OF DEVIATIONS

Any exceptions / deviations shall be reported to HOD (EM).

9.0 RECORD KEEPING:

Records will be maintained by concerned Area Maintenance Engineer in File: LMI/025.

10.0 REVIEW:

This LMI document will be reviewed once in every two years by Head of the Project, NTPC Seshadri.

CATHODIC PROTECTION READINGS Stage- 1/2				
LOCATION	SWEET WATER SYSTEM PIPELINE		SEA WATER SYSTEM PIPELINE	
PARAMETER	TR Unit (Raw Water P/H)	TR UNIT (Sweet Water P/H)	TR UNIT (EDP P/H)	TR UNIT (SEA WATER P/H 11 KV SWGR)
System Auto LD1:				
AC Input Voltage (M1):				
DC Output Voltage (M2):				
DC Output Current(M3):				
Corrosion Voltage(M4) (-)Ref. Electrode D5,LD6:				

TEST LEAD POINTS READINGS: Stage-1/2

SWEET WATER PIPE LINE (Date:) SEA WATER PIPE LINE(Date:)

Sl. No.	TLP No.	TLP (-) Voltage	Remarks	Sl.No.	TLP No.	TLP (-) Voltage	Remarks
1	TLP-1			1	TLP-1		
2	TLP-2			2	TLP-2		
3	TLP-3			3	TLP-3		
4	TLP-4			4	TLP-4		
5	TLP-5			5	TLP-5		
6	TLP-6			6	TLP-6		
7	TLP-7			7	TLP-7		
8	TLP-8			8	TLP-8		
9	TLP-9			9	TLP-9		
10	TLP-10			10	TLP-10		
11	TLP-11						
12	TLP-12						
13	TLP-13						
14	TLP-14						
15	TLP-15						
16	TLP-16						
17	TLP-17						
18	TLP-18						
NOTE: Required values (-)0850 to (-)1350 m Volts							

EMD Representative

OFS Representative

OPN Representative

DISTRIBUTION LIST:

1. HOD (EM)
2. INTRANET